

Problem Description:

Mr. Alex just installed new bands in his feet and they do not feel like they should, making it where he cannot stop!

However the Alex cannot do the math while stopping himself from crashing into things. Using $M = -dx$, please help him find out the M, d, or x value as he requests it.

Constraints:

M, D, X = 0 to 10000 (or a question mark).

Input Format:

Each line will start capital letters only, then the value of that variable. The value must be a decimal number.

Output Format:

Print the output in a separate lines contains the variable, a space, then the value.

Logical Test Cases

Test Case 1

INPUT (STDIN)

M 14.00
D -4.80
X ?

EXPECTED OUTPUT

x 2.92

Test Case 2

INPUT (STDIN)

M 12.00
D -5.80
X ?

EXPECTED OUTPUT

x 2.07

✓ Mandatory Test Cases

Test Case 1

KEYWORD

```
char var[3][LEN];
```

Test Case 2

KEYWORD

```
char inp[3][LEN];
```

✓ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

4

Test Case 2

TOKEN COUNT

320

Test Case 3

NLOC

30

Searching1.c +

```
1 #include <stdio.h>
2 #include <string.h>
3 #include <stdlib.h>
4
5 #define LEN 100
6
7 char var[3][LEN];
8 char inp[3][LEN];
9
10 int main() {
11     double m = 0, d = 0, x = 0;
12     int m_idx = -1, d_idx = -1, x_idx = -1;
13     for (int i = 0; i < 3; i++) {
14         if (scanf("%s %s", var[i], inp[i]) != 2) return 0;
15         if (strcmp(var[i], "M") == 0 || strcmp(var[i], "m") == 0) {
16             if (strcmp(inp[i], "?") != 0) m = atof(inp[i]);
17             else m_idx = i;
18         } else if (strcmp(var[i], "D") == 0 || strcmp(var[i], "d") == 0) {
19             if (strcmp(inp[i], "?") != 0) d = atof(inp[i]);
20
21         } else if (strcmp(var[i], "X") == 0 || strcmp(var[i], "x") == 0) {
22             if (strcmp(inp[i], "?") != 0) x = atof(inp[i]);
23             else x_idx = i;
24         }
25     }
```

main | CH.SC.U4CSE25225 | Searching1.c

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```
26  if (x_idx != -1) {
27      x = m / (-d);
28      printf("x %.2f\n", x);
29  } else if (d_idx != -1) {
30      d = -m / x;
31      printf("d %.2f\n", d);
32  } else if (m_idx != -1) {
```

```
33     m = -d * x;  
34     printf("m %.2f\n", m);  
35 }  
36 return 0;  
37}  
38
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

M 14.00

D -4.80

X ?

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching1.exe

x 2.92

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

M 12.00

D -5.80

X ?

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching1.exe

x 2.07

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

DS	Session	Searching	Question Information	Level 1	Challenge 2
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Problem Description:
King Alexander wants every chariot to line up for the start of her Winter-eve ball.

He has asked you, Twilight Sparkle, to sort the horse chariots alphabetically but with royalty in front. Royal horses chariot have diamonds in their names.

Constraints:
1 <= Names <= 100

Input Format:
list of horse chariots numbering anywhere from 1 to 40 horse chariot, with 1 per line. The end of input will be marked with END on a line by itself. Names should be no longer than 100 characters in length, and to only contain letters and spaces.

Output Format:
Print the output in a separate lines contains Sort and list the horse chariots alphabetically in ascending order ('A' "first"), ignoring case. However, any horse chariot with a diamond in their name must be placed at the "top" of the list (before the "A's" start) in the diamond order given in the Discussion section below.

Explanation:
The Pony ranking (in Ascending order) for gemstones is a follows:

Lapis, Topaz, Tourmaline, Sapphire, Peridot, Ruby, Pearl, Emerald, Diamond, Aquamarine, Amethyst, Garnet.

you are guaranteed that you will not have to deal with any gemstones not listed above. If multiple gems are listed in the same name, sort by whichever gem has "highest prioity" (e.g. a pony named 'Garnet Lapis' would be listed before a pony named 'Topaz Sapphire,' because Lapis has the highest priority).

In the case of equal ranking on gemstones, output in alpha order by total name (not just the gemstones), only sort by highest priority gemstones, after that by alpha. (Example, given the names: Lapis Topaz and Amethyst Lapis, they should be printed in order as: Amethyst Lapis then Lapis Topaz. That is because both names have the highest ranked gemstone (Lapis), so we simply sort them in ascending order alphabetically after their ranking has been established. We would not list Lapis Topaz before Amethyst Lapis because Topaz has a higher ranking over Amethyst. Stop comparing gemstone ranking after determine the highest rank of the gems in the name.)

You are guaranteed that there will be no ties in priority for gemstone ranking. You are also guaranteed that you will not encounter hyphenated names like Ruby-Sue. However, if a gemstone name happened to be found as part of a name (like Rubyanne) you can safely treat that as just another name to alphabetise. Gemstone names have to stand on their own

(separated by spaces, or the entirety of the name) to be treated royally.

Logical Test Cases

Test Case 1

INPUT (STDIN)

Buttershy
Orangejack
Rubyanne
Rainbow Dash
Rarity
Pinkie Pie
Emerald Sunshine
Spike the Dragon
Misty Sapphire
END

EXPECTED OUTPUT

Misty Sapphire
Emerald Sunshine
Buttershy
Orangejack
Pinkie Pie
Rainbow Dash
Rarity
Rubyanne
Spike the Dragon

Test Case 2

INPUT (STDIN)

Granny Smith
Discord
Spitfire
Granny Smith
Photo Finish
Pearl Peridot
Granny Smith
Princess Pearl
Soarine
END

EXPECTED OUTPUT

Pearl Peridot
Princess Pearl
Discord
Granny Smith
Granny Smith
Granny Smith
Photo Finish
Soarine
Spitfire

Mandatory Test Cases

Test Case 1

KEYWORD

char *gems[]=
{ "NONE", "Garnet", "Amethyst", "Aquamarine", "Diamond", "Emerald", "Pearl", "Ruby", "Peridot", "Sapphire", "Tourmaline", "Topaz", "Lapis", 0 };

Test Case 2

KEYWORD

char ponies [MAXP] [BUFLN];

Test Case 3

KEYWORD

strcmp(ponies[a],ponies[b])>0;

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

4

Test Case 2

TOKEN COUNT

530

Test Case 3

NLOC

60

```
Searching2.c +
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4
5 #define MAXP 1000
6 #define BUFLen 100
7
8 char ponies[MAXP][BUFLen];
9 char *gems[] = {
10     "NONE", "Garnet", "Amethyst", "Aquamarine", "Diamond", "Emerald", "Pearl",
11     "Ruby", "Peridot", "Sapphire", "Tourmaline", "Topaz", "Lapis", 0
12};
13
14int get_gem_rank(char *name) {
15    int max_rank = 0;
16    char temp[BUFLen];
17    strcpy(temp, name);
18    char *token = strtok(temp, " ");
19    while (token != NULL) {
20        for (int i = 1; gems[i] != 0; i++) {
21            if (strcmp(token, gems[i]) == 0) {
22                if (i > max_rank) {
23                    max_rank = i;
24                }
25            }
26        }
27    }
28}
```

```

26     }
27     token = strtok(NULL, " ");
28 }
29 return max_rank;
30}
31
32int main() {
33     int n = 0;
34     char line[BUFLen];
35     while (fgets(line, sizeof(line), stdin)) {
36         line[strcspn(line, "\r\n")] = 0;
37         if (strlen(line) == 0) continue;
38         if (strcmp(line, "END") == 0) break;

```

main | CH.SC.U4CSE25225 | Searching2.c

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```

40     }
41
42     int ranks[MAXP];

```

```
43  for (int i = 0; i < n; i++) {
44      ranks[i] = get_gem_rank(ponies[i]);
45  }
46
47  for (int i = 0; i < n - 1; i++) {
48      for (int j = 0; j < n - i - 1; j++) {
49          int a = j;
50          int b = j + 1;
51          int swap = 0;
52          if (ranks[a] < ranks[b]) {
53              swap = 1;
54          } else if (ranks[a] == ranks[b]) {
55              if (strcmp(ponies[a], ponies[b]) > 0) {
56                  swap = 1;
57              }
58          }
59          if (swap) {
```

```
60         char tname[BUFLen];
61         strcpy(tname, ponies[a]);
62         strcpy(ponies[a], ponies[b]);
63         strcpy(ponies[b], tname);
64         int tr = ranks[a];
65         ranks[a] = ranks[b];
66         ranks[b] = tr;
67     }
68 }
69 }
70
71 for (int i = 0; i < n; i++) {
72     printf("%s\n", ponies[i]);
73 }
74 return 0;
75}
76
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

Buttershy
Orangejack
Rubyanne
Rainbow Dash
Rarity
Pinkie Pie
Emerald Sunshine
Spike the Dragon
Misty Sapphire

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching2.exe

Misty Sapphire
Emerald Sunshine
Buttershy
Orangejack
Pinkie Pie
Rainbow Dash
Rarity
Rubyanne
Spike the Dragon

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

Granny Smith
Discord
Spitfire
Granny Smith
Photo Finish
Pearl Peridot
Granny Smith

Princess Pearl

Soarine

END

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching2.exe

Pearl Peridot

Princess Pearl

Discord

Granny Smith

Granny Smith

Granny Smith

Photo Finish

Soarine

Spitfire

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

DS	Session	Searching	Question Information	Level 1	Challenge 3
Problem Description: Darsh recently found an new rectangular circuit board that he would like to recycle. That has R rows and C columns of squares. Each square of the circuit board has a very thinness, measured in millimetres. The square in the r-th row and c-th column has thinness $M_{r,c}$. A circuit board is beautiful if in each row, the difference between the thinnest square and the least thin square is no greater than L. Since the original circuit board might not be beautiful, Darsh would like to find a beautiful subcircuit board. This board can be obtained by choosing an axis-aligned subrectangle from the original board and taking the squares in that subrectangle. Darsh would like your help in finding the number of squares in the largest beautiful subrectangle of his original board. Constraints: $1 \leq T \leq 50$. $1 \leq R \leq 205$. $1 \leq C \leq 205$. $0 \leq M_i, j \leq 10^3$ for all i, j. $0 \leq L \leq 10^3$. Input Format: The first line of the input gives the number of test cases, T. T test cases follow. Each test case begins with one line containing three integers R, C and L, the number of rows, the number of columns, and the maximum difference in thinness allowed in each row. Then, there are R more lines containing C integers each. The c-th integer on the r-th line is M_r, c, the thinness of the square in the r-th row and c-th column. Output Format: Print the output in a separate lines contains the maximum number of squares in a beautiful subrectangle.					

Logical Test Cases

Test Case 1

INPUT (STDIN)

3
1 4 0
3 1 3 3
2 3 0
4 4 5
7 6 6
4 5 0
2 2 4 4 20
8 3 3 3 12
6 6 3 3 3
1 6 8 6 4

EXPECTED OUTPUT

2
2
6

Test Case 2

INPUT (STDIN)

3
2 4 0
2 3 1 2
3 1 3 3
2 3 0
41 4 5
7 61 6
4 5 0
12 12 14 4 20
81 13 13 3 12
61 16 31 3 3
11 16 81 6 4

EXPECTED OUTPUT

2
2
4

Mandatory Test Cases

Test Case 1

KEYWORD

int A[309][309];

Test Case 2

KEYWORD

bool ok[309][309][309];

Complexity Test Cases

Test Case 1

Test Case 2

Test Case 3

CYCLOMATIC COMPLEXITY

13

TOKEN COUNT

400

NLOC

53

```
Searching3.c +
1 #include <stdio.h>
2 #include <stdbool.h>
3
4 int A[309][309];
5 bool ok[309][309][309];
6
7 int main() {
8     int T;
9     if (scanf("%d", &T) != 1) return 0;
10    while (T--) {
11        int R, C, K;
12        scanf("%d %d %d", &R, &C, &K);
13        for (int i = 0; i < R; i++) {
14            for (int j = 0; j < C; j++) {
15                scanf("%d", &A[i][j]);
16            }
17        }
18        for (int i = 0; i < R; i++) {
19            for (int c1 = 0; c1 < C; c1++) {
20                int mn = A[i][c1];
21                int mx = A[i][c1];
22                for (int c2 = c1; c2 < C; c2++) {
23                    if (A[i][c2] < mn) mn = A[i][c2];
24                    if (A[i][c2] > mx) mx = A[i][c2];
25                    if (mx - mn <= K) {
```

```
26         ok[i][c1][c2] = true;
27     } else {
28         ok[i][c1][c2] = false,
29     }
30 }
31 }
32 }
33 int max_area = 0;
34 for (int c1 = 0; c1 < C; c1++) {
35     for (int c2 = c1; c2 < C; c2++) {
```

main | CH.SC.U4CSE25225 | Searching3.c

Ln 53, Col 1 Spaces: 4 UTF-8 CRLF C

```
36     int width = c2 - c1 + 1;
37     int current_h = 0;
38     for (int r = 0; r < R; r++) {
39         if (ok[r][c1][c2]) {
40             current_h++;
41             int area = current_h * width;
42             if (area > max_area) max_area = area;
43         } else {
44             current_h = 0;
```

```
45         }
46     }
47 }
48 }
49     printf("%d\n", max_area);
50 }
51 return 0;
52}
53
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
3
```

```
1 4 0
```

```
3 1 3 3
```

```
2 3 0
```

```
4 4 5
```

```
7 6 6
```

```
4 5 0
```

```
2 2 4 4 20
```

```
8 3 3 3 12
```

```
6 6 3 3 3
```

```
1 6 8 6 4
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching3.exe
```

```
2
```

```
2
```

```
6
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
3
```

```
2 4 0
```

```
2 3 1 2
```

```
3 1 3 3
```

```
2 3 0
```

```
41 4 5
```

```
7 61 6
```

```
4 5 0
```

```
12 12 14 4 20
```

```
81 13 13 3 12
```

```
61 16 31 3 3
```

11 16 81 6 4

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching3.exe

2

2

4

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

DS	Session	Searching	Question Information	Level 1	Challenge 4
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Question Description:

Ragu has given a range [L, R] to Smith. Smith wants to require to find the number of integers 'X' in the range such that $GCD(X, F(X)) > 1$ where F(x) is equal to the sum of digits of 'X' in its hexadecimal (or base 16) representation.

Example : $F(27) = 1+B=1+11=12$

(27 in hexadecimal is written as 1B)

Constraints:

$$1 \leq T \leq 50$$

$$1 \leq L$$

$$R \leq 10^5$$

Input Format:

The first line contains a positive integer 'T' denoting the number of questions that you are asked.

Each of the next 'T' lines contain two integers 'L' and 'R' denoting the range of questions.

Output Format:

Print the output in a separate lines exactly 'T' numbers as the output.

Sample Input:

3
1 3
5 8
7 12

Sample output:

2
4
6

Explanation

For the first test-case, numbers which satisfy the criteria specified are : 2,3. Hence, the answer is 2.

For the second test-case, numbers which satisfy the criteria specified are : 5,6,7 and 8. Hence, the answer is 4.

For the third test-case, numbers which satisfy the criteria specified are : 7,8,9,10,11 and 12. Hence, the answer is 6.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

3
1 3
5 8
7 12

Test Case 2

INPUT (STDIN)

10
15 64
14 100
9 79
19 63
3 83
5 61
18 93

EXPECTED OUTPUT

2
4
6

20 98
13 71
12 53

EXPECTED OUTPUT

29
52
45
26
53
37
45
47
36
26

✓ Mandatory Test Cases

Test Case 1

KEYWORD

`int search(int a, int b)`

✓ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

165

Test Case 3

NLOC

27

Searching4.c +

```
1 #include <stdio.h>
2
3 int gcd(int a, int b) {
4     while (b) {
5         int t = a % b;
6         a = b;
7         b = t;
8     }
9     return a;
10}
11
12int f_hex(int x) {
13    int sum = 0;
14    while (x > 0) {
15        sum += (x % 16);
16        x /= 16;
17    }
18    return sum;
19}
20
21
22 int count = 0;
23 for (int x = a; x <= b; x++) {
24     if (gcd(x, f_hex(x)) > 1) {
25         count++;
```

main | CH.SC.U4CSE25225 | Searching4.c

Ln 40, Col 1 Spaces: 4 UTF-8 CRLF C

```
26     }
27 }
28 return count;
29}
30
31int main() {
32     int t, l, r;
33     if (scanf("%d", &t) != 1) return 0;
```

```
34 while (t--) {  
35     scanf("%d %d", &l, &r);  
36     printf("%d\n", search(l, r));  
37 }  
38 return 0;  
39}  
40
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

3

1 3

5 8

7 12

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching4.exe

2

4

6

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

10

15 64

14 100

9 79

19 63

3 83

5 61

18 93

20 98

13 71

12 53

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching4.exe

29

52

45

26

53

37

45

47

36

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

DS	Session	Searching	Question Information	Level 1	Challenge 5
Problem Description: Nesamani wants to paint a beautiful picture on a building that is M sections long. Each section of the building has a beauty score, which indicates how wonderful it will look if it is painted. Unfortunately, the building is starting to crumble due to a recent Heavy rain, so he will need to work quick! At the beginning of each day, Nesamani will paint one of the sections of the building. On the first day, he is free to paint any section he likes. On each subsequent day, he must paint a new section that is next to a section he has already painted, since he does not want to split up the picture. At the end of each day, one section of the building will be destroyed. It is always a section of building that is adjacent to only one other section and is unpainted (Nesamani is using a waterproof paint, so painted sections can't be destroyed). The total beauty of Nesamani's picture will be equal to the sum of the beauty scores of the sections he has painted. Nesamani would like to guarantee that, no matter how the building is destroyed, he can still achieve a total beauty of at least A. What's the maximum value of A for which he can make this guarantee? Constraints: $1 \leq T \leq 100$. $2 \leq M \leq 100$. Input Format: The first line of the input gives the number of test cases, T. Each test case starts with a line containing an integer M. Then, another line follows containing a string of M digits from 0 to 9. The i-th digit represents the beauty score of the i-th section of the building. Output Format: Print the output in a separate lines contains the maximum beauty score that Nesamani can guarantee that he can achieve, as described above.					

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

4
4
1332
4
9583
3
616
10
1029384756

EXPECTED OUTPUT

6
14
7
31

Test Case 2

INPUT (STDIN)

4
3
332
3
583
4
1616
8
29384756

EXPECTED OUTPUT

6
13
7
24

▼ Mandatory Test Cases

Test Case 1

KEYWORD

for(k=1;k<=T;++k)

Test Case 2

KEYWORD

vector<int> b(N+1);

▼ Complexity Test Cases

Test Case 1

Test Case 2

Test Case 3

CYCLOMATIC COMPLEXITY

4

TOKEN COUNT

175

NLOC

25

Searching5.c +

```
1 #include <stdio.h>
2 #include <string.h>
3
4 #if 0
5 vector<int> b(N+1);
6 #endif
7
8 int main() {
9     int T, k;
10    if (scanf("%d", &T) != 1) return 0;
11    for (k = 1; k <= T; ++k) {
12        int N;
13        scanf("%d", &N);
14        char s[10005];
15        scanf("%s", s);
16        int L = (N + 1) / 2;
17
18        for (int i = 0; i < L; i++) {
19            sum += s[i] - '0';
20        }
21        int max_sum = sum;
22        for (int i = L; i < N; i++) {
23            sum += (s[i] - '0') - (s[i - L] - '0');
24            if (sum > max_sum) {
25                max_sum = sum;
```

main | CH.SC.U4CSE25225 | Searching5.c

Ln 32, Col 1 Spaces: 4 UTF-8 CRLF C

```
26     }  
27 }  
28 printf("%d\n", max_sum);  
29 }
```

```
30     return 0;  
31}  
32
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

4

4

1332

4

9583

3

616

10

1029384756

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching5.exe

6

14

7

31

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

4

3

332

3

583

4

1616

8

29384756

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching5.exe

6

13


```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

DS	Session	Searching	Question Information	Level 1	Challenge 6
Problem Description: Kanna is upset to learn that no one at his school recognises his first name. Even his friends refer to him by his surname. Frustrated, he decides to make his fellow college students know his first name by forcing them to solve this question. The task is determining the third greatest number in the supplied array. Constraints: $0 < n < 100$ $0 < arr < 1000$ Input Format: first line represents the number of elements N to be get second line indicates input elements according to N Output Format: Single line represents the out put that is third largest number.					

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

```
6
1 14 2 16 10 20
```

EXPECTED OUTPUT

The third Largest element is 14

Test Case 2

INPUT (STDIN)

```
7
19 -10 20 14 2 16 10
```

EXPECTED OUTPUT

The third Largest element is 16

▼ Mandatory Test Cases

Test Case 1

KEYWORD

```
void thirdLargest(int
arr[],int arr_size)
```

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

7

Test Case 2

TOKEN COUNT

220

Test Case 3

NLOC

44

Searching6.c +

```
1 #include <stdio.h>
2 #include <limits.h>
3
4 void thirdLargest(int arr[], int arr_size) {
5     if (arr_size < 3) {
6         return;
7     }
8     int first = INT_MIN, second = INT_MIN, third = INT_MIN;
9     for (int i = 0; i < arr_size; i++) {
10         if (arr[i] > first) {
11             third = second;
12             second = first;
13             first = arr[i];
14         } else if (arr[i] > second && arr[i] != first) {
15             third = second;
16             second = arr[i];
17         } else if (arr[i] > third && arr[i] != second && arr[i] != first) {
18
19         }
20     }
21     printf("The third Largest element is %d\n", third);
22 }
23
24 int main() {
25     int n;
```

main | CH.SC.U4CSE25225 | Searching6.c

Ln 34, Col 1 Spaces: 4 UTF-8 CRLF C

```
26     if (scanf("%d", &n) != 1) return 0;
27     int arr[1000];
28     for (int i = 0; i < n; i++) {
29         scanf("%d", &arr[i]);
30     }
```

```
31  thirdLargest(arr, n);  
32  return 0;  
33}  
34
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
6
```

```
1 14 2 16 10 20
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching6.exe
```

```
The third Largest element is 14
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
7
```

```
19 -10 20 14 2 16 10
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching6.exe
```

```
The third Largest element is 16
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

Question Description:

Suresh have "N" rectangles.

A rectangle is Silver if the ratio of its sides is in between $[1.6, 1.7]$, both inclusive. Your task is to find the number of silver rectangles.

Constraints:

$$1 \leq N \leq 10^5$$

$$1 \leq W,$$

$$H \leq 10^9$$

Input Format:

First line: Integer "N" denoting the number of rectangles Each of the "N" following lines:

Two integers W, H denoting the width and height of a rectangle

Output Format:

Print the output in a single line contains find the number of Silver rectangles.

Sample Input:

5

10 1

165 100

180 100

170 100

160 100

Sample Output:

3

Explanation:

There are three Silver rectangles: (165, 100), (170, 100), (160, 100).

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

4
185 100
180 100
170 100
160 100

EXPECTED OUTPUT

2

Test Case 2

INPUT (STDIN)

6
185 100
180 100
170 100
160 100
165 90
164 95

EXPECTED OUTPUT

2

▼ Mandatory Test Cases

Test Case 1

Test Case 2

KEYWORD

```
if(width/height>=1.6 &&  
width/height<=1.7)
```

KEYWORD

```
else if(height/width >=1.6  
&& height/width<=1.7)
```

✓ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

7

Test Case 2

TOKEN COUNT

125

Test Case 3

NLOC

25

Searching7.c +

```
1 #include <stdio.h>
2
3 int main() {
4     int n;
5     if (scanf("%d", &n) != 1) return 0;
6     int count = 0;
7     for (int i = 0; i < n; i++) {
8         double width, height;
9         scanf("%lf %lf", &width, &height);
10        if (width / height >= 1.6 && width / height <= 1.7) {
11            count++;
12        } else if (height / width >= 1.6 && height / width <= 1.7) {
13            count++;
14        }
15    }
16    printf("%d\n", count);
17    return 0;
18}
19
```

Ln 19, Col 1 Spaces: 4 UTF-8 CRLF C

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

4

185 100

180 100

170 100

160 100

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching7.exe

2

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

6

185 100

180 100

170 100

160 100

165 90

164 95

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching7.exe

2

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

DS	Session	Searching	Question Information	Level 1 Challenge 8
----	---------	-----------	----------------------	-----------------------

Problem Description:

Ram's first job is a out of university is for a student's book publisher. They are currently publishing a line of "intelligent libs" books, and they need you to check their print-copy. Write a program that will take a sentence with placeholders in it, and substitute in for those placeholders words from the lists of example words the publisher has provided.

Your goal will be to substitute the appropriate words into the sentence from the list of provided words TWICE (keeping track of the words you have used) where:

- [N] = nouns
- [AV] = adverbs
- [V] = verbs
- [A] = adjectives

Input Format:

Line one will always be the line with the story. The order of the Nouns, Adverbs, Verbs and Adjectives will always be the same. Input stops when it reaches the word END alone on a line. You are guaranteed to always have enough words to substitute into the sentence TWICE (not repeating any words). The number of words given under each section will be random.

Output Format:

Print the output in a separate lines contains, Always take the TOP-most UNUSED word from a group when substituting. Output the final sentences when all substitutions are done.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

There once was a [AJ] [N] from [N] who [AV] [V] all day.
NOUNS
squirrel
London
tree
lettuce
ADVERBS
silently
quickly

Test Case 2

INPUT (STDIN)

There once was a [AJ] [N] from [N] who [AV] [V] all day.
NOUNS
singer
chennai
hotel
chettinad
ADVERBS
silently
quickly

happily
VERBS
skipped
ran
climbs
ADJECTIVES
delightful
homely
quain

EXPECTED OUTPUT

There once was a delightful squirrel from London who
silently skipped all day.
There once was a homely tree from lettuce who quickly ran
all day.

happily
sadly
VERBS
skipped
reads
ran
sings
climbs
ADJECTIVES
delightful
homely
surprisingly

EXPECTED OUTPUT

There once was a delightful singer from chennai who silently
skipped all day.
There once was a homely hotel from chettinad who quickly
reads all day.

▼ Mandatory Test Cases

Test Case 1

KEYWORD

int cl[CMDS]

Test Case 2

KEYWORD

char *lists[CMDS][MAXWORDS];

Test Case 3

KEYWORD

char *tokens[TOKENS]={"[N]","[AV]","[V]","[AJ]"};

Test Case 4

KEYWORD

char *cmds[CMDS]={
"NOUNS","ADVERBS","VERBS","ADJECTIVE
S","END"};

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

3

Test Case 2

TOKEN COUNT

488

Test Case 3

NLOC

50

Searching8.c +

```
1 #include <stdio.h>
2 #include <string.h>
3 #include <stdlib.h>
4
5 #define CMDS 5
6 #define MAXWORDS 100
7
8 void substitute(char *sentence, char *lists[CMDS][MAXWORDS], int *list_counts) {
9     int used[CMDS] = {0};
10
11     for (int pass = 0; pass < 2; pass++) {
12         char result[4000] = "";
13         char *ptr = sentence;
14
15         while (*ptr) {
16             if (*ptr == '[') {
17                 ptr++;
18                 char placeholder[10];
19                 int j = 0;
20                 while (*ptr && *ptr != ']') {
21                     placeholder[j++] = *ptr++;
22                 }
23                 placeholder[j] = '\\0';
24                 if (*ptr == ']') ptr++;
25
```

```

26     int cat = -1;
27     if (strcmp(placeholder, "N") == 0) cat = 0;
28     else if (strcmp(placeholder, "AV") == 0) cat = 1;
29     else if (strcmp(placeholder, "V") == 0) cat = 2;
30     else if (strcmp(placeholder, "AJ") == 0) cat = 3;
31
32     if (cat >= 0 && used[cat] < list_counts[cat]) {
33         strcat(result, lists[cat][used[cat]]);
34         used[cat]++;
35     } else {
36         strncat(result, "[", 1);
37         strncat(result, placeholder, strlen(placeholder));
38         strncat(result, "]", 1);
39     }
40 } else {
41     strncat(result, ptr, 1);
42     ptr++;
43 }
44 }
45

```

```

46     printf("%s\n", result);
47 }
48}
49
50int main() {
51     char line[4000];
52     char sentence[4000];
53     char *lists[CMDS][MAXWORDS];
54     int list_counts[CMDS];
55     char *cmds[CMDS] = {"NOUNS", "ADVERBS", "VERBS", "ADJECTIVES", "END"};
56
57     sentence[0] = '\0';
58
59     for (int i = 0; i < CMDS; i++) {
60         list_counts[i] = 0;
61         for (int j = 0; j < MAXWORDS; j++) {
62             lists[i][j] = NULL;

```

main | CH.SC.U4CSE25225 | Searching8.c

Ln 124, Col 1 Spaces: 4 UTF-8 CRLF C

```

64     }
65

```

```
66  int current_section = -1;
67
68  while (fgets(line, sizeof(line), stdin)) {
69      line[strcspn(line, "\n")] = 0;
70
71      if (strcmp(line, "END") == 0) {
72          if (sentence[0] != '\0') {
73              substitute(sentence, lists, list_counts);
74          }
75          break;
76      }
77
78      if (strchr(line, '[') != NULL) {
79          if (sentence[0] != '\0') {
80              substitute(sentence, lists, list_counts);
81
82              for (int i = 0; i < CMD5; i++) {
83                  for (int j = 0; j < list_counts[i]; j++) {
84                      free(lists[i][j]);
85                  }

```

```

86         }
87
88         for (int i = 0; i < CMDS; i++) {
89             list_counts[i] = 0;
90             for (int j = 0; j < MAXWORDS; j++) {
91                 lists[i][j] = NULL;
92             }
93         }
94     }
95
96     strcpy(sentence, line);
97     current_section = -1;
98     continue;
99 }
100
101 int is_header = -1;
102 for (int i = 0; i < CMDS - 1; i++) {
103     if (strcmp(line, cmds[i]) == 0) {
104         is_header = i;
105         break;

```

```

106     }
107 }
108
109 if (is_header >= 0) {
110     current_section = is_header;
111 } else if (current_section >= 0 && list_counts[current_section] < MAXWORDS) {
112     lists[current_section][list_counts[current_section]] = strdup(line);
113     list_counts[current_section]++;
114 }
115 }
116
117 for (int i = 0; i < CMDS; i++) {
118     for (int j = 0; j < list_counts[i]; j++) {
119         free(lists[i][j]);
120     }
121 }
122
123 return 0;
124 }

```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

There once was a [AJ] [N] from [N] who [AV] [V] all day.

NOUNS

squirrel

London

tree

lettuce

ADVERBS

silently

quickly

VERBS

skipped

ran

climbs

ADJECTIVES

delightful

homely

quain

END

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching8.exe

There once was a delightful squirrel from London who silently skipped all day.

There once was a homely tree from lettuce who quickly ran all day.

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

There once was a [AJ] [N] from [N] who [AV] [V] all day.

NOUNS

singer

chennai

hotel

chettinad

ADVERBS

silently

quickly

sadly

VERBS

skipped

reads

ran

sings

```
climbs
ADJECTIVES
delightful
homely
surprisingly
END
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching8.exe
There once was a delightful singer from chennai who silently skipped all day.
There once was a homely hotel from chettinad who quickly reads all day.
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

DS	Session	Searching	Question Information	Level 1	Challenge 9
----	---------	-----------	----------------------	---------	-------------

Problem Description:

Prabhu Salamon is planning to make a very long journey across the cityside by Train. His journey consists of N train routes, numbered from 1 to N in the order he must take them. The trains themselves are very fast, but do not run often. The i-th train route only runs every Xi days.

More specifically, he can only take the i-th train on day Xi, 2Xi, 3Xi and so on. Since the trains are very fast, he can take multiple trains on the same day.

Prabhu Salamon must finish his journey by day D, but he would like to start the journey as late as possible. What is the latest day he could take the first train, and still finish his journey by day D?

It is guaranteed that it is possible for Prabhu Salamon to finish his journey by day D.

Constraints:

- $1 \leq T \leq 100$.
- $1 \leq Xi \leq D$.
- $1 \leq N \leq 1000$.
- $1 \leq D \leq 10^{12}$

Input Format:

The first line of the input gives the number of test cases, T. T test cases follow. Each test case begins with a line containing the two integers N and D. Then, another line follows containing N integers, the i-th one is Xi.

Output Format:

Print the output in a single line contains, the latest day he could take the first train, and still finish his journey by day D.

Logical Test Cases

Test Case 1

INPUT (STDIN)

3
3 100

Test Case 2

INPUT (STDIN)

3
5 1000

```
31 71 21
4 100
11 16 51 50
1 1
1
```

EXPECTED OUTPUT

```
62
44
1
```

```
311 171 211 45 32
6 1000
111 161 511 510 56 93
2 10
9 5
```

EXPECTED OUTPUT

```
622
0
9
```

▼ Mandatory Test Cases

Test Case 1

KEYWORD

```
for(int t=0;t<T;t++)
```

Test Case 2

KEYWORD

```
for(int i=n-1;i>=0;i--)
```

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

4

Test Case 2

TOKEN COUNT

138

Test Case 3

NLOC

30

Searching9.c +

```
1 #include <stdio.h>
2
3 int main() {
4     int T;
5     if (scanf("%d", &T) != 1) return 0;
6     for (int t = 0; t < T; t++) {
7         int n;
8         long long d;
9         scanf("%d %lld", &n, &d);
10        long long x[1005];
11        for (int i = 0; i < n; i++) {
12
13            }
14            long long cur = d;
15            for (int i = n - 1; i >= 0; i--) {
16                cur = (cur / x[i]) * x[i];
17            }
18            printf("%lld\n", cur);
19        }
20        return 0;
21    }
22
```

main | CH.SC.U4CSE25225 | Searching9.c

Ln 22, Col 1 Spaces: 4 UTF-8 CRLF C

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

3

3 100

31 71 21

4 100

11 16 51 50

1 1

1

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching9.exe

62

44

1

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

3

5 1000

311 171 211 45 32

6 1000

111 161 511 510 56 93

2 10

9 5

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching9.exe

622

0

9

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

DS	Session	Searching	Question Information	Level 1	Challenge 10
Question Description: An array A contains integers with the following constraints: 'A' contains elements in sorted order. Integer i occurs $i \cdot \text{floor}(\sqrt{i}) + \text{ceil}(i/2)$ times in the array. All elements are greater than or equal to 1. Fathima has given Q queries of type: L R: Find the number of distinct values present in subarray A[L...R]. Note: 1-based indexing is followed. Constraints: $1 \leq Q \leq 10^5$ $1 \leq L \leq R \leq 10^{13}$ Input Format: The first line contains an integer Q denoting the number of queries. Next Q lines contains two space-separated integers L, R, denoting the query. Output Format: Print the output in separate lines contains required number of distinct values. Example Sample Input: 2 1 3 1 6 Sample Output: 2					

Explanation

First few elements of the array A are 1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, ...

- For Query 1:-
 - Number of distinct elements in subarray $A[1...3]$ is 2.
- For Query 2:-
 - Number of distinct elements in subarray $A[1...6]$ is 3.

Logical Test Cases

Test Case 1

INPUT (STDIN)

2
3 3
2 6

EXPECTED OUTPUT

1
3

Test Case 2

INPUT (STDIN)

3
3 3
2 6
5 4

EXPECTED OUTPUT

1
3
1

Mandatory Test Cases

Test Case 1

KEYWORD

while(l<ans1)

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

6

Test Case 2

TOKEN COUNT

400

Test Case 3

NLOC

56

Searching10.c +

```
1 #include <stdio.h>
2
3 long long get_sum(long long n) {
4     long long sum_i = n * (n + 1) / 2;
5     long long sum_ceil;
6     if (n % 2 == 0) {
7         long long k = n / 2;
8         sum_ceil = k * (k + 1);
9     } else {
10         long long k = (n + 1) / 2;
11         sum_ceil = k * k;
12     }
13     return sum_i + sum_ceil;
14}
15
16 long long get_val(long long pos) {
17     long long low = 1, high = 50000000;
18     long long ans = high;
19     while (low <= high) {
20         long long mid = low + (high - low) / 2;
21         if (get_sum(mid) >= pos) {
22             ans = mid;
23             high = mid - 1;
24         } else {
```

```
26     }
27 }
28 return ans;
29}
30
31int main() {
32     int q;
33     if (scanf("%d", &q) != 1) return 0;
34     while (q--) {
35         long long l, r;
36         scanf("%lld %lld", &l, &r);
37         if (l > r) {
```

```
38     long long tmp = 1;
39     l = r;
40     r = tmp;
41 }
42 long long v1 = get_val(l);
43 long long v2 = get_val(r);
44 printf("%lld\n", v2 - v1 + 1);
45 }
46 return 0;
47}
48
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

2

3 3

2 6

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching10.exe

1

3

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

3

3 3

2 6

5 4

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Searching10.exe

1

3

1

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>